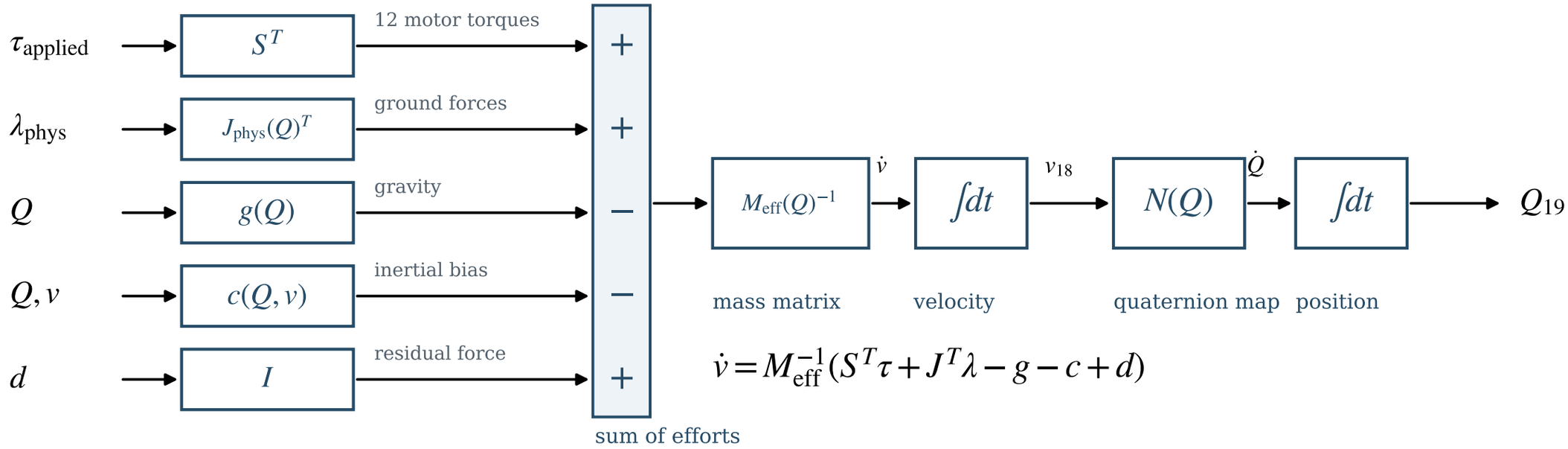
V16 | Physical plant built from small blocks

18-velocity-coordinate PhysX plant. Physical contact force comes from the contact solver, not the controller QP prediction.

EFFORT SOURCES

STATE EVOLUTION



$$\dot{v} = M_{\text{eff}}^{-1}(S^T\tau + J^T\lambda - g - c + d)$$

WHAT EACH BLOCK MEANS

$S^T\tau = [0_6; \tau_{12}]$ no base motor	$J^T\lambda$ foot force to generalized effort	M_{rigid} 54 masses, COMs, inertias	rotor armature 0.02 kg m^2 per joint

$$M_{\text{eff}} = M_{\text{rigid}} + S^T(0.02I_{12})S, \quad h = c + g, \quad \dot{Q} = N(Q)v$$

State-dependent M, J, g, c and N read the current Q, v; the outer state return is on page 1.

Actual USD joint damping is zero. d includes unmodeled joint friction. Contact forces can be distributed; J^T lambda is their generalized-force reduction.